

Scalable Azo-dye Based Carbon Quantum Dot For Optical Management

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01 INTRODUCTION

Azo dyes are a versatile class of organic compounds widely recognized for their intense coloration, photophysical properties, and broad applications in industrial and biomedical fields. The current investigation explores their potential as carbon dot precursors through a facile solvothermal synthesis approach for optoelectronic applications.

PL EMISSION 486nm **EL Emission** 565nm **CIE COORDINATES** (0.27, 0.32)

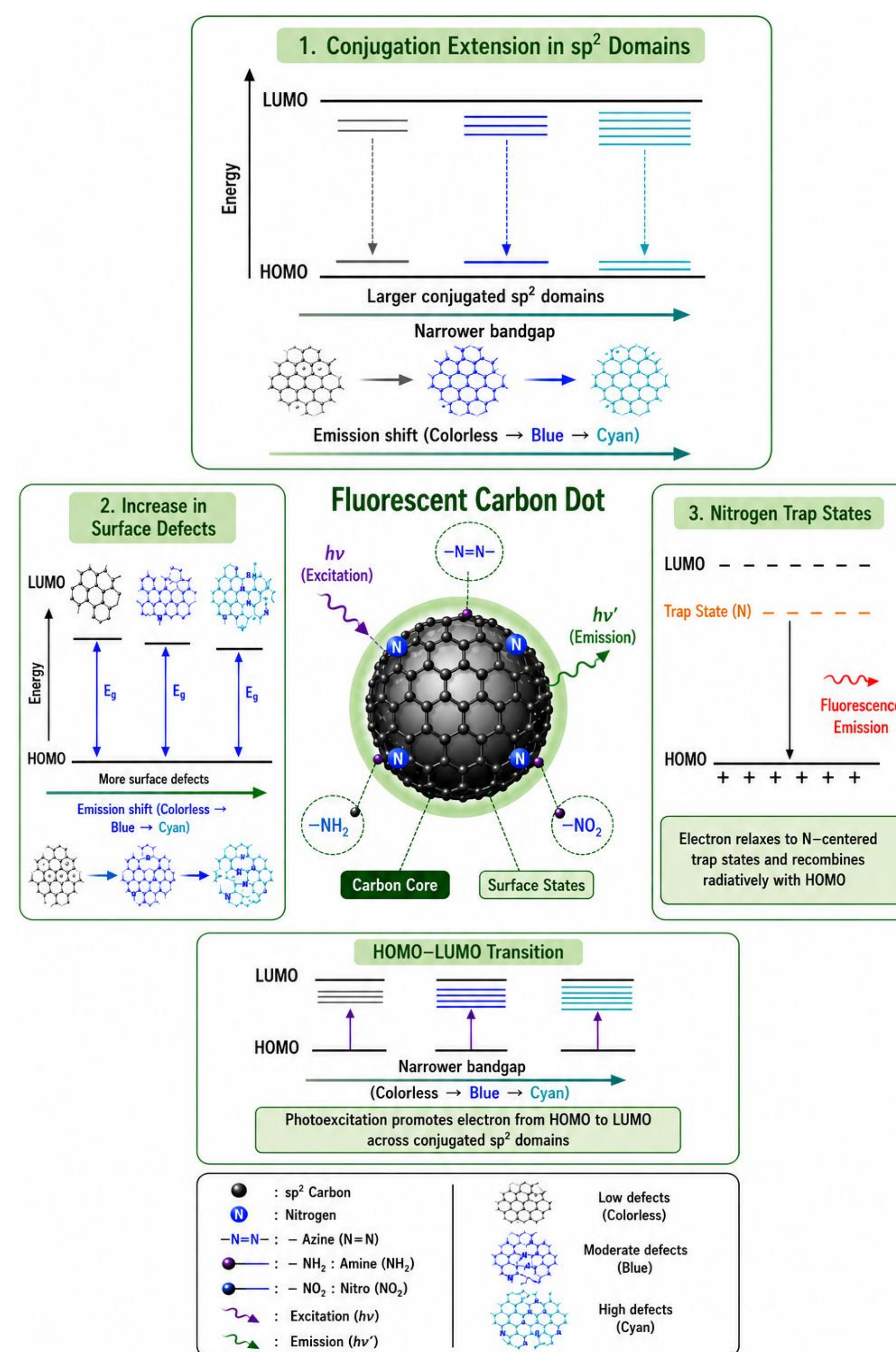
02 OBJECTIVES

- ✓ To synthesize N-Carbon Quantum Dots (N-CQDs) from an azo dye precursor using a facile one-pot solvothermal approach.
- ✓ To characterize the structural, morphological, and optical properties of the synthesized CQDs and fabricate a PMMA-based luminescent down-conversion LED.
- ✓ To demonstrate the potential of azo dye-derived CQDs for applications in light-emitting devices, sensing, and advanced photonic technologies.

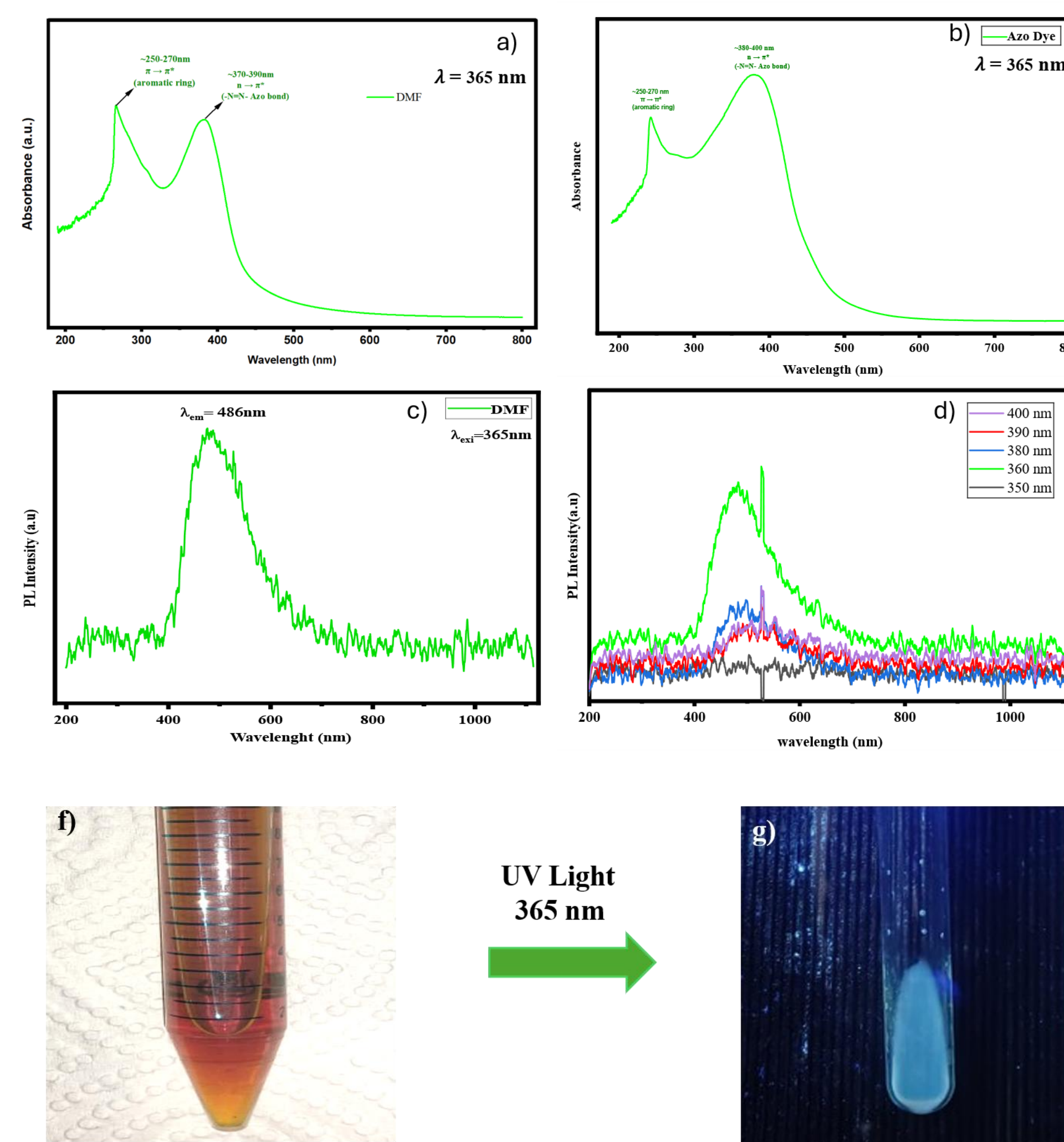
03. SYSTHESIS OF AZO-CQD



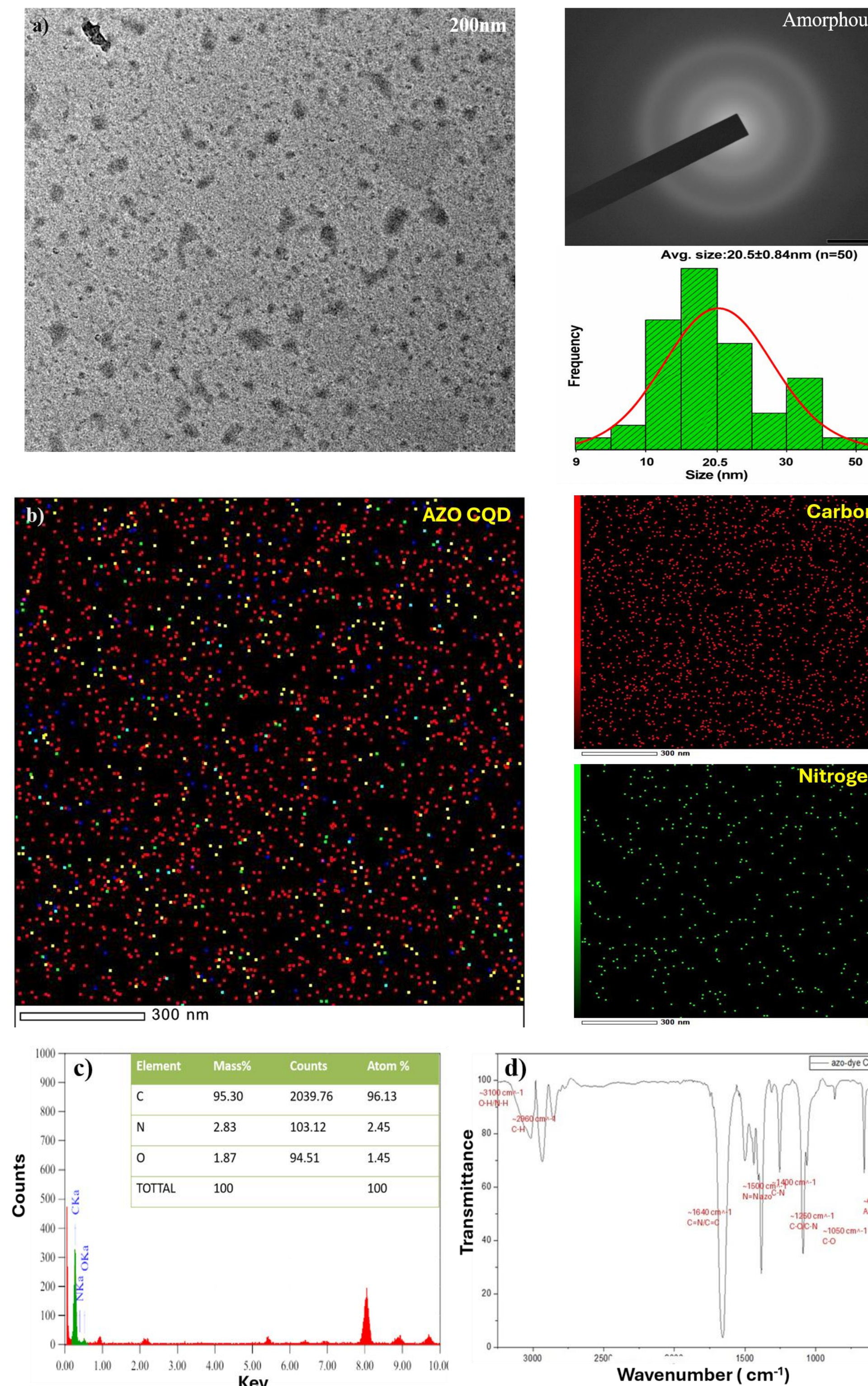
04 MECHANISM



05 OPTICAL PROPERTIES

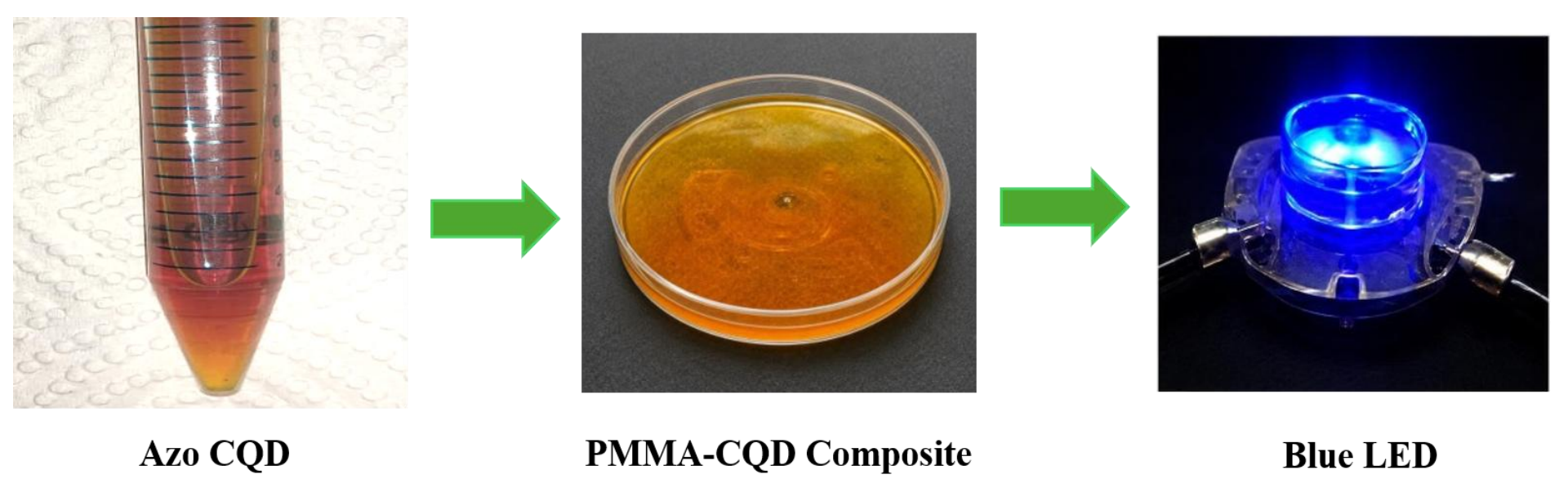


06 CHARACTERIZATIONS

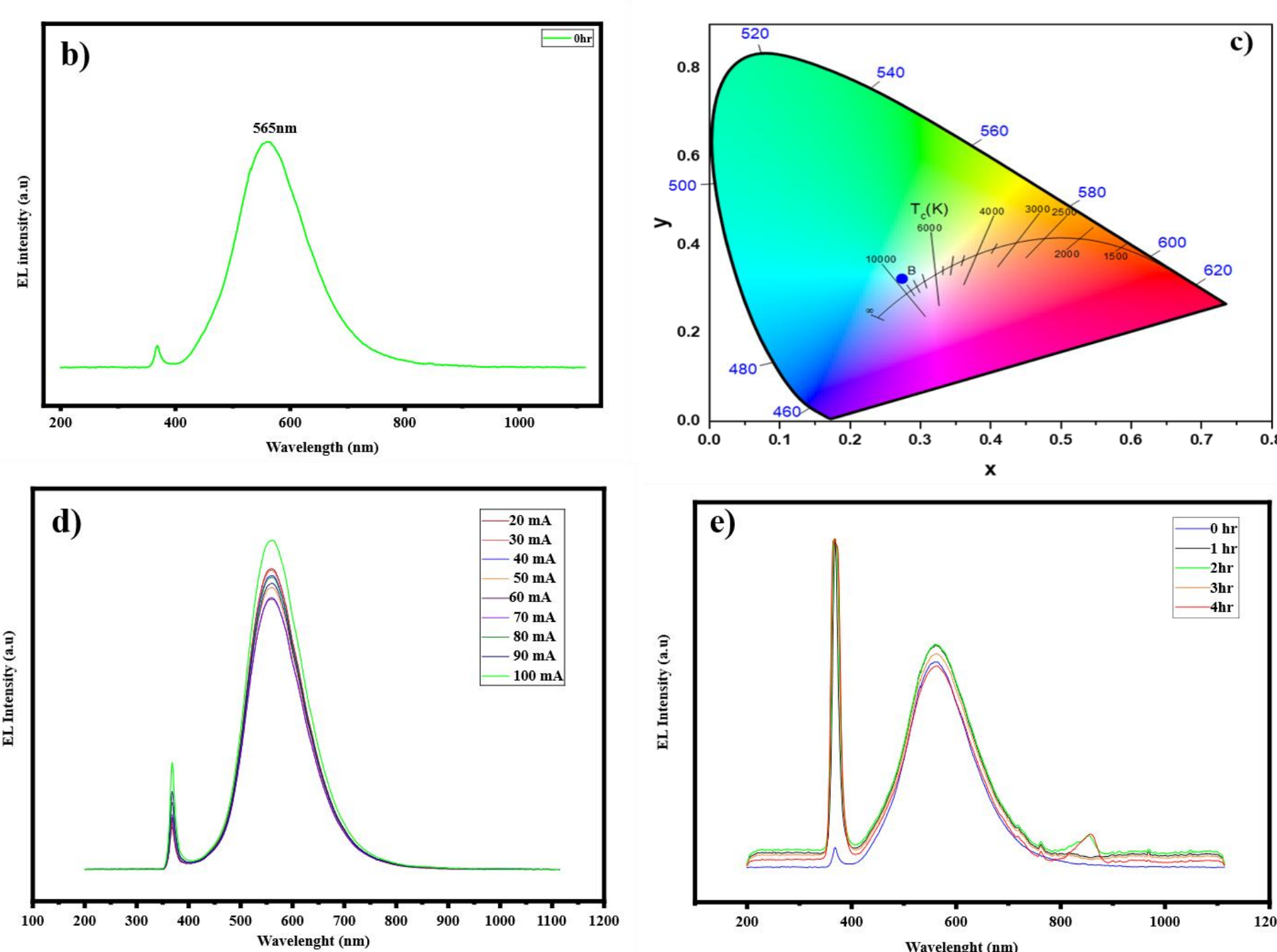


07 OPTICAL APPLICATION

a) Fabrication of blue LED



Composite was made by adding few drops of AZO CQD into the PMMA dissolved in DMF. Once DMF is evaporated in the oven for few days the below Electroluminescence study was carried out



Efficient blue-to-green down-conversion demonstrates the potential of azo dye-derived CQDs for energy-efficient solid-state lighting

06 CONCLUSIONS

- Azo dyes were successfully converted into highly fluorescent material
- PMMA-based down-conversion LED, demonstrating their suitability for optoelectronic applications
- The successful conversion of azo dyes into highly emissive carbon quantum dots provides a sustainable and versatile platform for optoelectronic, sensing, and photonic applications.

07 FUTURE WORK

Our future research will explore the same underlying synthesis mechanism to develop red-emissive carbon dots for advanced optoelectronic devices and biomedical imaging.

08 REFERENCES

- Guo, H., Liu, Z., Shen, X., & Wang, L. (2022). One-pot synthesis of orange emissive carbon quantum dots for all-type high color rendering index white light-emitting diodes. *ACS Sustainable Chemistry & Engineering*, 10(26), 8289-8296.

09 ACKNOWLEDGEMENTS

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